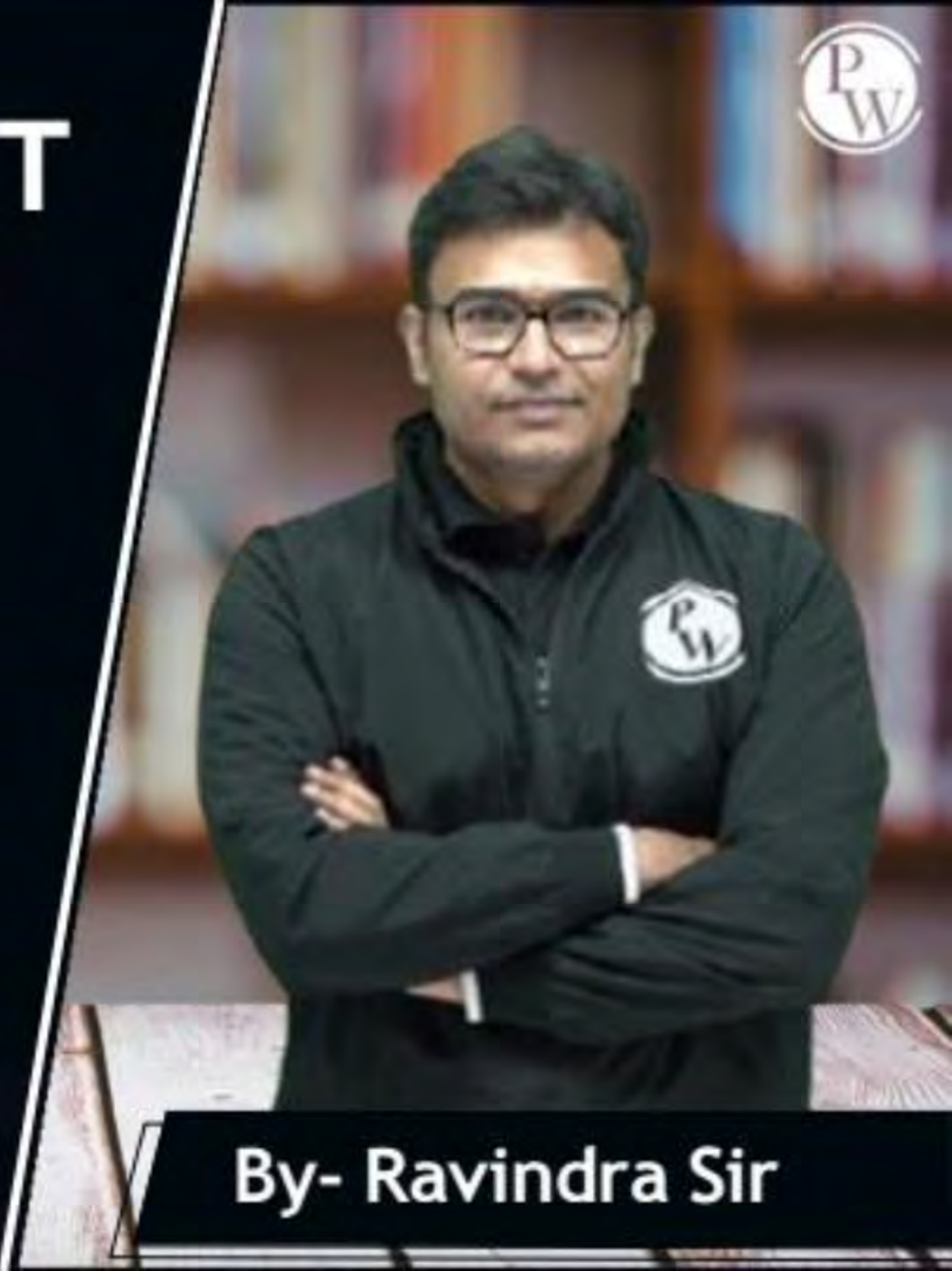
Computer Science & IT

COMPUTER NETWORK
(CN)

ISO-OSI Stack

Lecture No. 04



By- Ravindra Sir

Recap of Previous Lecture



Topic

Topic

Topic

Switching

Topics to be Covered



- Topic — IPV4
- Topic
- Topic

Inspiring Stories : Abdul Khader Nadakattin



Background: Farmer from Karnataka, loved solving farm problems.

Education: No big degree, self-taught.

Achievements: Made machines for tamarind seed removal, sowing, tilling, and more.

Impact: Saved farmers' time and effort across villages.

Inspiring Stories : Ramar Pillai



Background: Village boy from Tamil Nadu, obsessed with fuel.

Education: School dropout.

Achievements: Claimed to make herbal fuel from plants.

Impact: Sparked new ideas for cheap local energy.

Inspiring Stories : Sonam Wangchuk



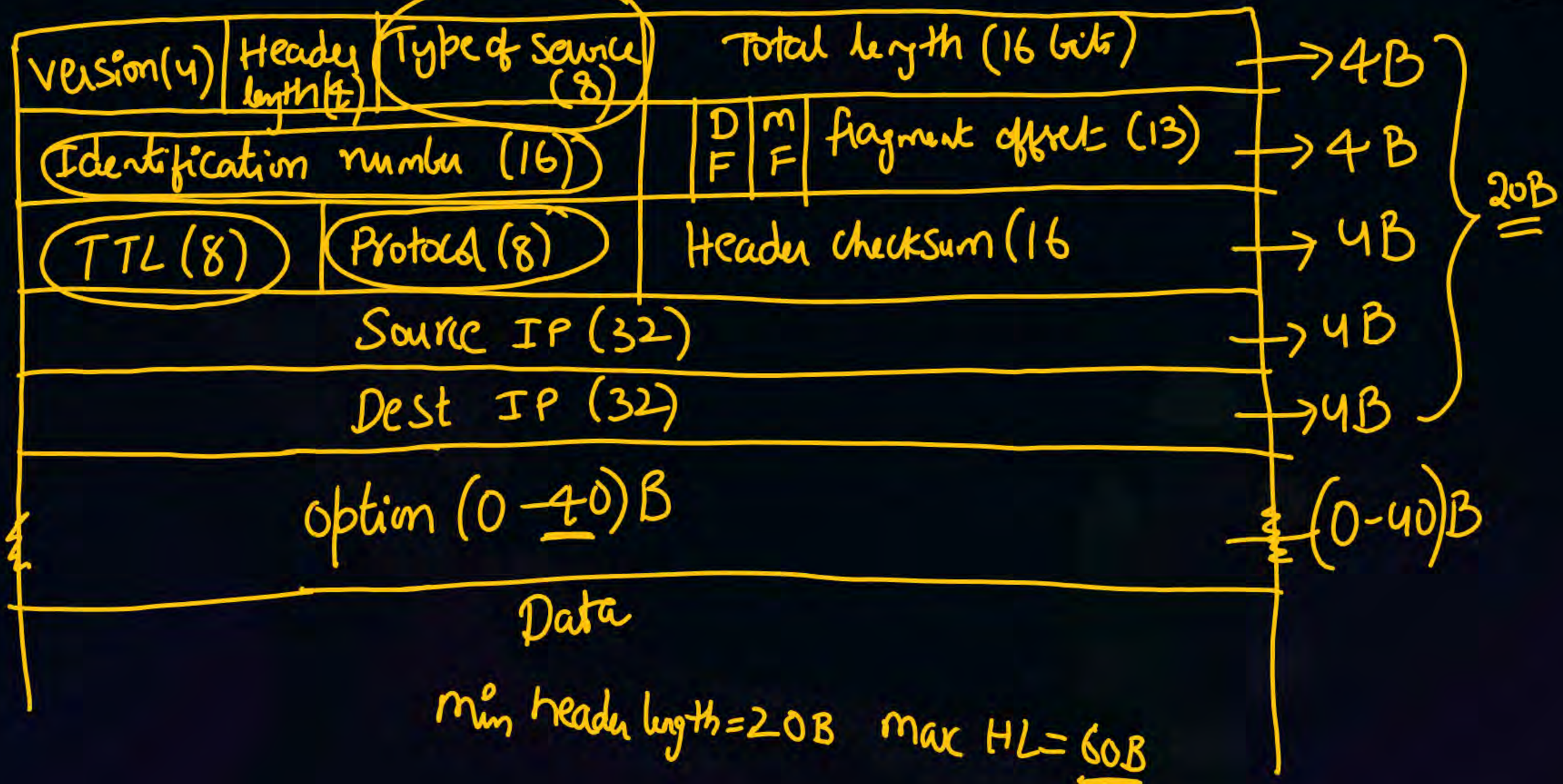
Background: From Ladakh, struggled with poor schools.

Education: Engineer, NIT Srinagar.

Achievements: Made Ice Stupas (artificial glaciers) and reformed Ladakhi education.

Impact: Helped farmers with water and inspired climate solutions.

IPV4 header

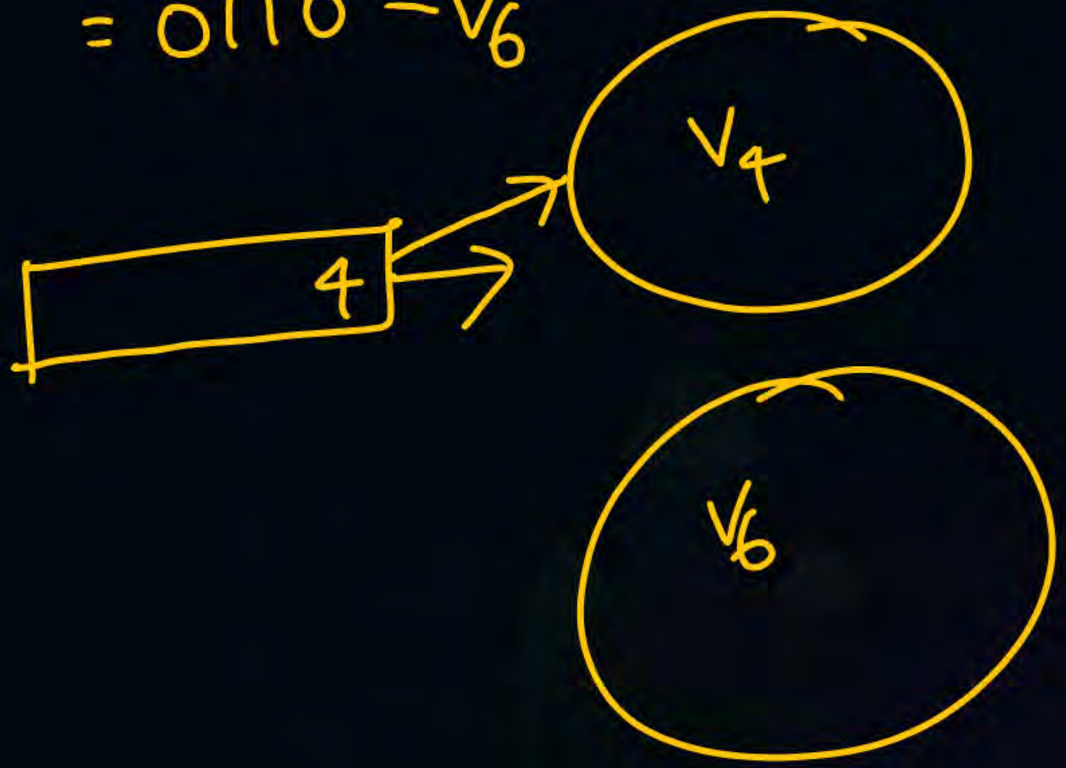


Min header length = 20B max HL = 60B

Version (4 bits)

V1
V2
V3
V_4 → we are at V_4
V_5 → discarded
V_6 → we are using it

Version = 0100 − V_4
= 0110 − V_6



Header length :- (20B - 60B)

Header length field = 4 bits 4 bits - 1111 - 15

Scaling factor - $\frac{60B}{15B} = 4$ ✓

HL	HLF
20B	5
32B	8
40B	10
60B	15

HL (30) + 2 padding = 32B

HLF $30/4 = 7.5$ ✓ $\lceil 7.5 \rceil = 8$ ✓

$HL = 8 \times 4 = 32$

11 — 32B — 8

Whenever header length is not a multiple of 4, we use padding in options we use a scaling factor of 4 in the header length field.

Identification number (16 bits) :-

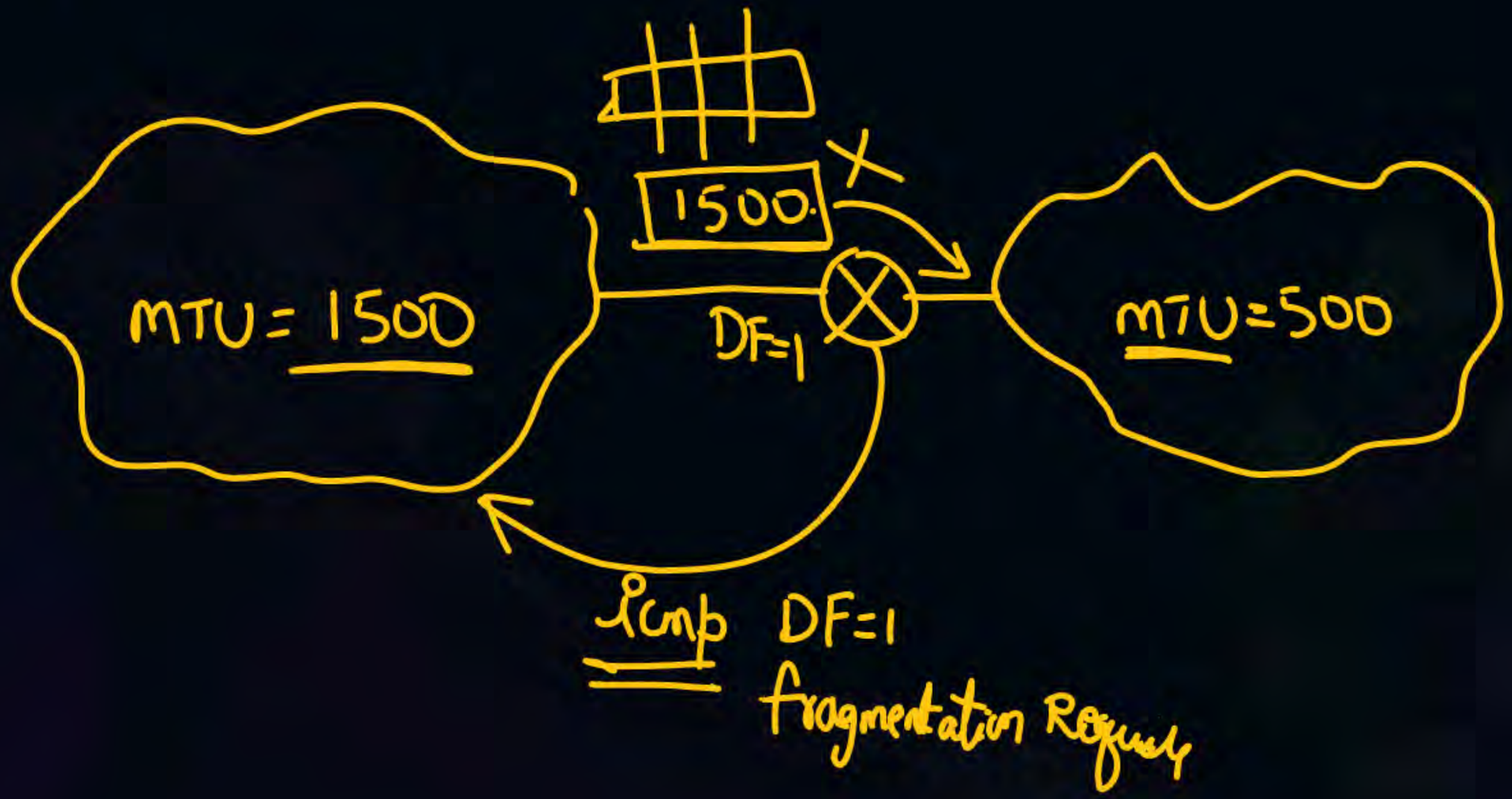
Every datagram coming out of a host will get an identification num

when we do fragmentation, all the fragments will have same ID number



DF :- Dont fragment

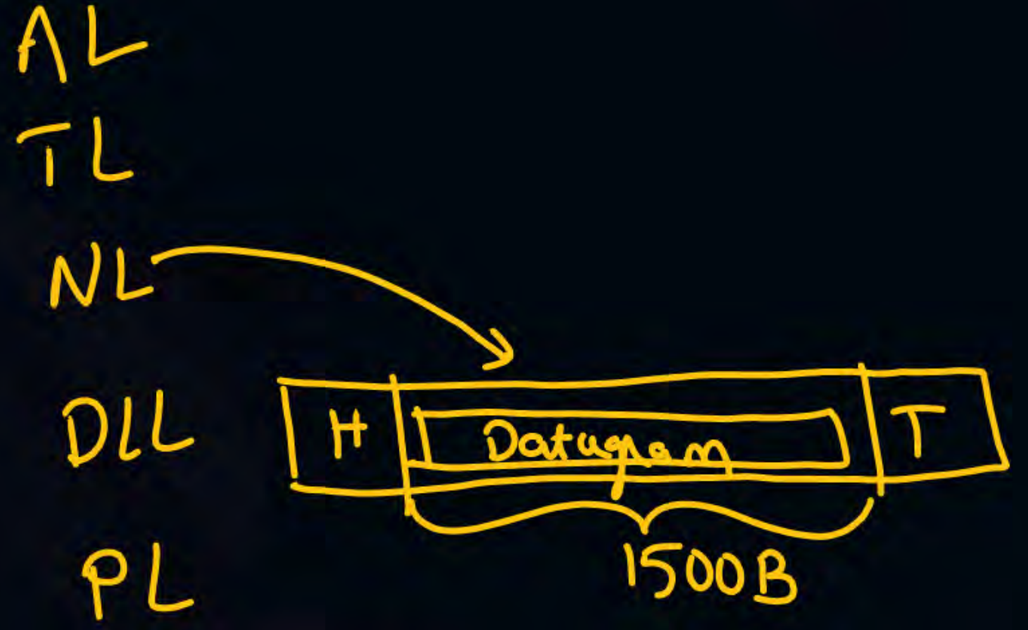
if DF=1, then router will not fragment the packet



MTU-
maximum transmittable unit

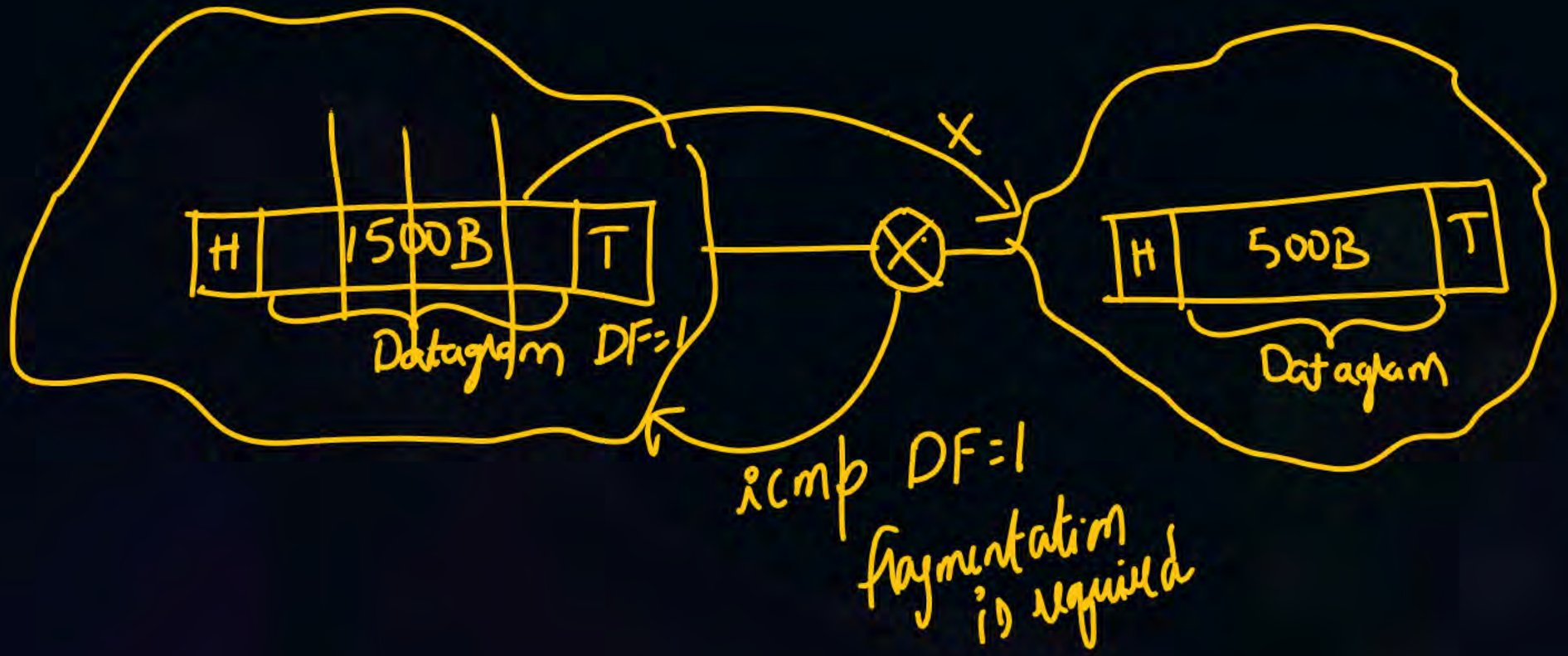
DLL - frame





MTU = 1500B

MTU :- The maximum size of a datagram that a DLL frame can carry.



MF :- more fragments are following

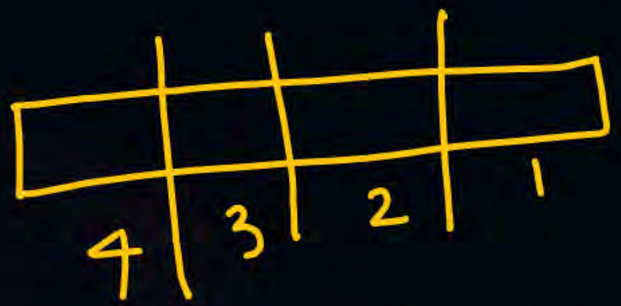


Datagram

When a datagram is fragmented, all the fragments except the last fragment will get MF = 1

Flagment offset (13 bits):-

It is used to know the order of fragments



Rec

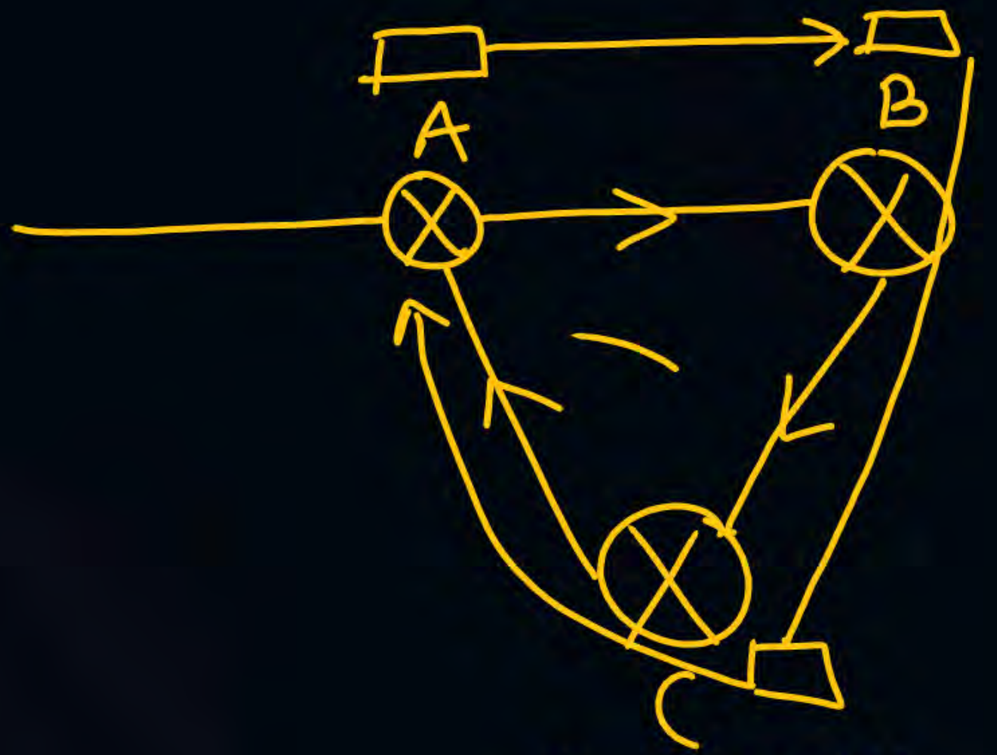


We reassemble a datagram which is fragmented.

Reassemble will happen at rec.

In order to reassemble, a datagram we use flagment offset.

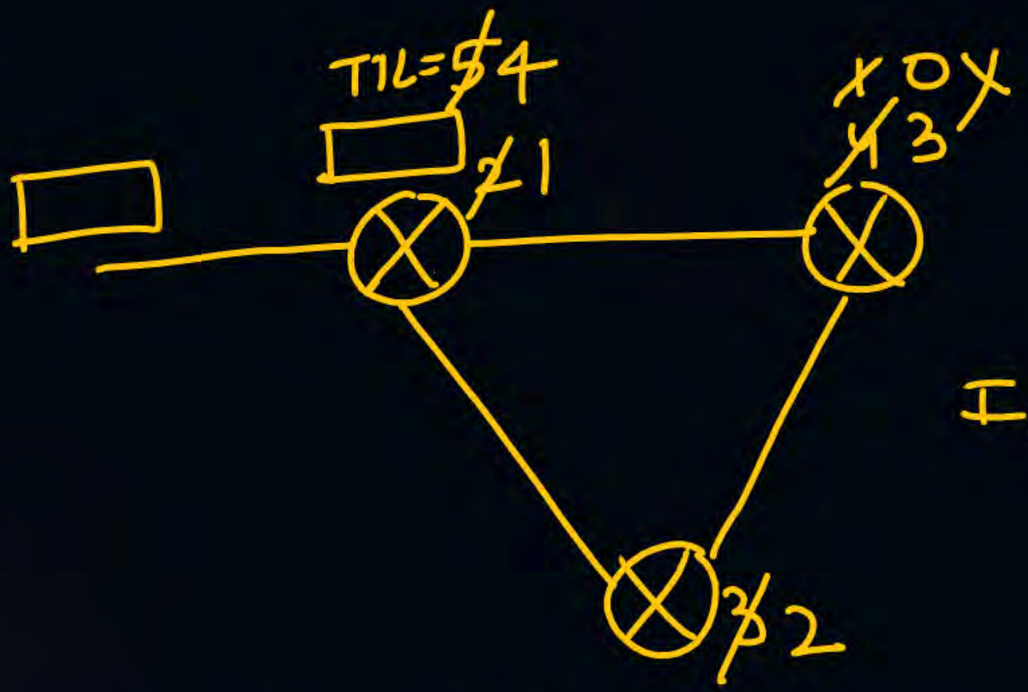
Time to live

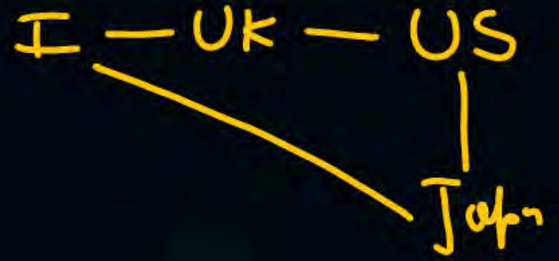


BW wasted

CPU wasted

Buffer get full



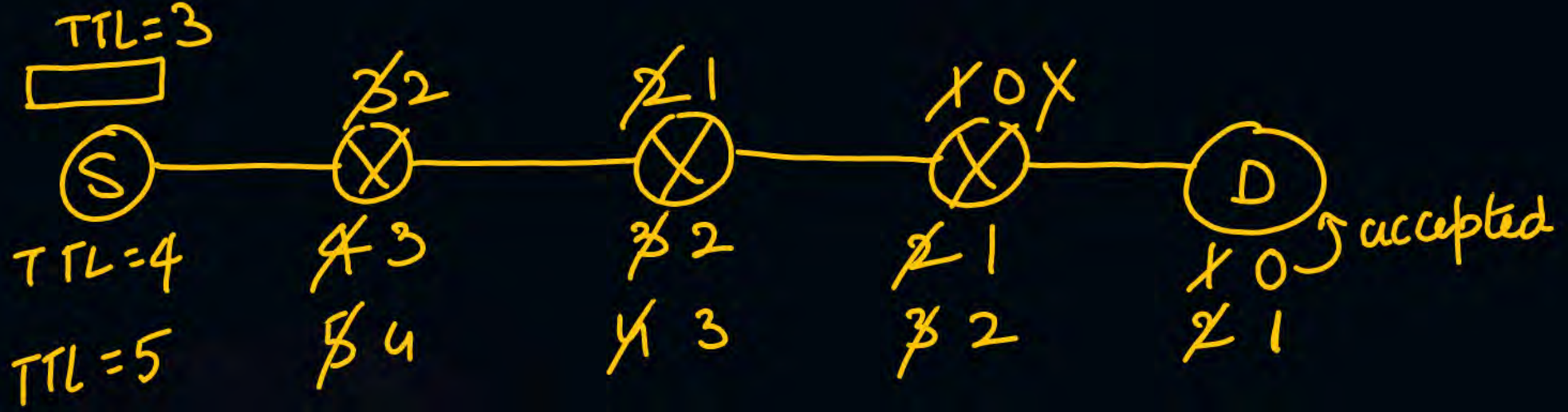


TTL is used to prevent a datagram from infinite looping

TTL is configurable.

16 ✓

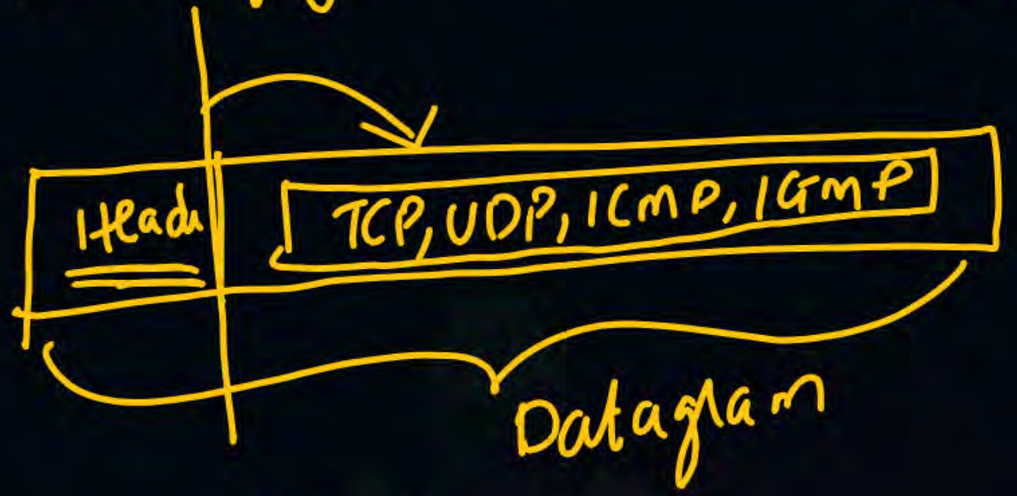
8 ✓


TTL=3
S
TTL=4
TTL=5
2
3
4
1
2
3
1
0
1
2
1
D
0
1
accepted

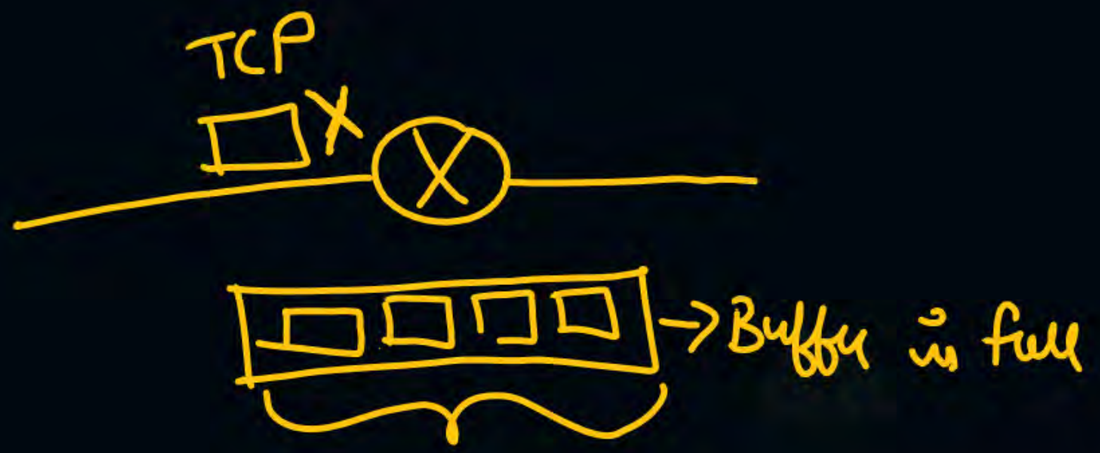


Protocol:-

Protocol field tells if a datagram is carrying ICMP & IGMP & TCP & UDP



NL ✓
DLL
PL
⊗


TCP
X
X
→ Buffer is full



× ICMP (LP) ✓

↓

× IGMP ×

↓

× UDP ✓

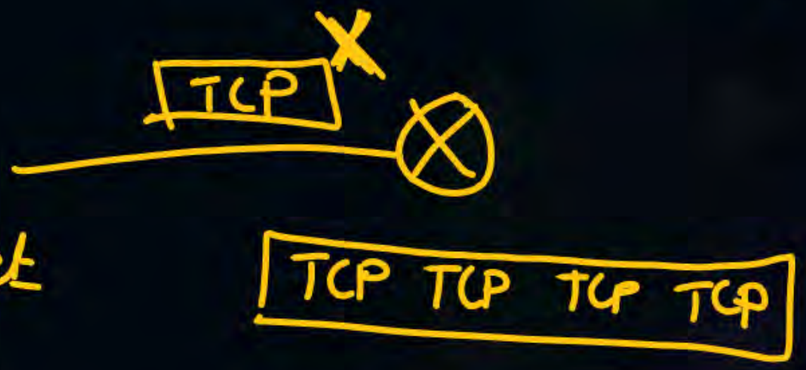
↓

× TCP (HP) ✓

TCP has highest priority

because if we discard a TCP packet it will retransmitted again





Header checksum :-

THANK - YOU